

Preference-Based Assessments

An Acquired Taste: Latent Class Analysis to Compare Adolescent and Adult Preferences for EQ-5D-Y-3L Health States

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ABSTRACT

Objectives: US stakeholders advised including adolescents in the valuation study for the EQ-5D-Y-3L, a step toward greater acknowledgment of children in informing societal values. This study aimed to assess the relative contribution of adolescent and adult preferences to a model when combined.

Methods: Discrete choice experiment data were collected from an online sample of 1529 adults and 714 adolescents (ages 11–17). Each respondent completed 15 discrete choice experiment tasks, which were analyzed using latent class models representing varying number of preference classes. Within the best-fitting model, the contribution of each class was determined by the scale-adjusted class share (SACS), combining the class's proportion of respondents (class share) and the magnitude of coefficients (within-class scale). We estimated the contribution of adolescent and adult respondents to SACS for each class, with lower SACS representing less contribution to the combined model.

Results: The best-fitting model described 6 classes. Adults had higher contribution to all except 1 class, accounting for 78.7% of the total contribution. After adjusting for the unequal sample size of adolescent and adult respondents, adults contributed approximately 65.0%, and adolescents contributed 35.0% of the weights toward a combined model.

Conclusions: Adolescents showed diminished, disproportionate representation in a combined model, due in part to more indifferent, less informative preferences for EQ-5D-Y-3L health states compared with adults. Latent class analysis showcases one approach to estimate and weight contributions from intentionally sampled subgroups in a combined model.

Keywords: discrete choice experiment, health-related quality of life, latent class analysis.

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Highlights

- There is interest in engaging children and adolescents in valuation of preference-based measures of health. However, there is uncertainty around the informativeness of adolescent preferences, particularly when combined with adult preferences.
- Through a latent class analysis of US adult and adolescent preferences for EQ-5D-Y-3L health states, we found that adolescents provided less informative preferences compared with adults. In a combined model with adults, adolescent contribution was disproportionately lower.
- The findings offer additional points of consideration for stakeholders and researchers advocating for the inclusion of adolescents in preference-based studies of health.

Introduction

Introduced in 2009, the EQ-5D-Y-3L is a generic measure of health designed for younger populations.¹ Similar to the EQ-5D-3L for adults, the EQ-5D-Y-3L descriptive system consists of 5 dimensions of health: mobility, looking after myself, doing usual activities, having pain or discomfort, and feeling worried, sad, or unhappy. Three response levels are used to describe the severity of problems in each dimension: no (or not), some (or a little), a lot (or very), and designated as levels 1, 2, and 3, respectively. Since the publication of an international valuation protocol in 2020, investigators worldwide have conducted valuation studies to estimate national value sets for the 243 health states described by the EQ-5D-Y-3L, providing utility scores for use in economic evaluations.^{2–11}

The international valuation protocol for the EQ-5D-Y-3L recommends eliciting preferences from adult respondents via discrete choice experiments (DCE) and composite time trade-off (cTTO) tasks, in which tasks are framed from the perspective of a 10-year-old child, ie, “Considering your views about a

10-year-old child, which do you prefer?”² Both DCE and cTTO are widely used methods in health economics to elicit preferences for health states and have been recommended as valuation tasks for EQ-5D instruments.¹² Compared with cTTO, which includes an element of duration, DCE without duration is a less cognitively demanding elicitation method, contributing to its popularity in valuing children's health-related quality of life (HRQoL), including directly from children and adolescents.^{13–17} In the international valuation protocol for the EQ-5D-Y-3L, DCE and cTTO methods are complementary when producing a value set.² DCE responses reflect preferences for the relative importance of attributes and response levels on an unanchored latent scale, whereas cTTO responses provide a basis for anchoring latent-scale DCE values to the 0 (dead) to 1 (full health) scale required for quality-adjusted life-year estimation.

Before valuation of the EQ-5D-Y-3L in the United States, investigators proactively engaged a panel of 14 national stakeholders to obtain additional guidance on adapting the recommendations in the international protocol for the US context.¹⁸ Although there was support from some stakeholders for only

including adults, consistent with eliciting national values from taxpayers, there was a lack of consensus from this diverse panel of US stakeholders in terms of who should be asked to value children's health states. To date, few studies have included children under 18 in societal preference studies. Based on arguments of inclusivity and respecting children's autonomy for making decisions about matters that affect their own health, other US stakeholders recommended eliciting preferences directly from adolescents in addition to adults.¹⁹ Considering the issues surrounding the framing perspectives used in valuation tasks, ethical arguments for and against the inclusion of adolescents in valuation and how adult and adolescent preferences for EQ-5D-Y-3L health states can differ,²⁰⁻²⁶ most stakeholders expressed support for including older children (ie, adolescents ages 11-17) who would be capable of completing valuation tasks for themselves.¹⁸ US adults and adolescents may have different preferences for dimensions of health and it could provide distinct, valuable information to intentionally elicit preferences from both.¹⁴ However, distinct subgroups' preferences in a model may be disproportionately represented when considering differences in the heterogeneity of their preferences from DCE data. Thus, we sought to explore an analytic approach to examine preferences from 2 different groups—adolescents and adults—to explore the connection between preference heterogeneity and the ultimate contribution from each group to a model.

Beyond the conditional multinomial logit approach that has been traditionally used to model choice behaviors, alternative models that can relax assumptions related to homogeneity of preferences and account for unobserved preference heterogeneity, such as mixed logit or latent class models, are increasingly used in health preference research.^{27,28} An ISPOR task force survey of 70 health preference researchers found that 58% of researchers agreed that all studies should aim to account for underlying preference heterogeneity because this leads to less biased coefficients and improved model fit.²⁹ Further, a complementary systematic review of DCE used in health preference research found that 110 studies (32%) used latent class analysis (LCA) as the method to account for preference heterogeneity.

LCA has been used for modeling preference data for value sets based on the EQ-5D-5L and EQ-5D-Y-3L.^{5,11,30} LCA is a robust and flexible modeling approach that can reveal how respondents cluster or form a discrete class, based on unobservable variation in innate preferences for relative importance of attributes (ie, taste heterogeneity), and the strength—or informativity—of those preferences (ie, scale heterogeneity).^{27,30} For example, certain respondents may place greater emphasis on minimizing problems with pain/discomfort over all other dimensions, resulting in coefficients with larger magnitude for level transitions within this dimension.^{31,32} In LCA, summing coefficients of a class can further characterize classes with strong preferences (larger magnitude of coefficients) or more indifferent preferences (smaller magnitude of coefficients). For example, respondents may demonstrate weak preferences across dimensions or levels, resulting in a class with universally small magnitudes of coefficients, typically referred to as a garbage class. Responses may lack meaningful indications of preferences when there is a probability of selecting one health state over another near 50%, regardless of the differences in the health-state descriptions.

A latent class (LC) model reflects each class weighted by the proportion of respondents that class represents (class share) and the relative scale of their coefficients. LCA is an optimal approach to reveal underlying taste and scale heterogeneity among adult and adolescent respondents through differences in their level of representation within latent classes and the individual magnitude of their modeled coefficients. In the context of combining

adolescent and adult responses, LCA offers an approach to quantitatively examine the relative contribution of adolescents and adults to a combined model. The contribution of adolescents and adults to the LC model is driven by the proportion of respondents in each class. The scales of coefficients for adults and adolescents are influenced by the preferences they indicate. Weak, indifferent preferences are ultimately less informative, resulting in smaller coefficients and therefore diminished in a model estimating values for health states. Uninformative preferences may be more likely obtained from respondents inexperienced with health states described by an instrument or inattentive when engaging with preference elicitation tasks, failing to represent the informed community-based preferences that are desirable to inform utility values for health technology assessment.^{33,34} If, for example, adolescents were to contribute less informative preferences compared with adults, the LC model would be skewed toward greater representation of adult responses disproportionate to their included sample size. Thus, in examining adult and adolescent preferences, it is first important to know how their prioritization of dimensions and informativity of their underlying preferences for EQ-5D-Y-3L health states may differ.

The specific objectives of this study were to explore LCA as a methodological approach to examining the latent preference heterogeneity of adolescent and adult respondents and to determine their relative contribution of information to a combined model.

Methods

Sampling and Recruitment

The target sample included a sample of approximately 1500 adults (ages ≥ 18 years) and 700 adolescents (ages 11 to 17 years) from an online panel survey company (Dynata) that includes participants from consumer brand loyalty programs regularly contacted for market research. Dynata contacted select quotas of respondents in the United States based on age, gender, race, ethnicity, and state of residence to attempt to achieve a diverse sample reflecting the composition of the general US population. All surveys were presented using the EuroQol Valuation Technology platform.¹² Participants successfully completing the survey without quality control (QC) violations were compensated for their participation in accordance with the terms of Dynata. Ethics approval was obtained from the Institutional Review Board at the University of San Francisco.

Design

Based on the international valuation protocol and previous EQ-5D-Y-3L valuation studies, the DCE experimental design consisted of 10 blocks with 15 different experimental pairs of health states and 3 fixed pairs for QC purposes (see [Appendix 2](#) in [Supplemental Materials](#) found at <https://doi.org/10.1016/j.jval.2025.01.020>).^{2,14} This design was chosen based on maximization of D-efficiency criteria without specifying priors. Respondents were randomly assigned to complete 1 block of tasks. For each task, the order and left-right positioning of health state pairs were randomly presented. In each pair of health states, 2 dimensions overlapped in their level of severity. The remaining 3 dimensions had different levels of severity and were presented with bolded font to help distinguish differences between health states. Tasks for adolescents were framed from their own perspective (which health state do you prefer?), whereas the tasks for adults were framed considering the perspective of a 10-year-old child (which health state do you prefer for a 10-year-old child?).² Respondents

were also asked to report their own health using the EQ-5D-Y-3L, complete background questions to describe their sociodemographic characteristics, and provide feedback to assess understanding of the DCE tasks.

Quality Control

Every respondent completed the same 3 DCE pairs (as the first pair, last pair, and randomly inserted among the 15 experimental pairs) for QC purposes. These tasks featured a dominated health state with more severe health problems for each dimension compared with the alternative. Responses were assessed for data quality by defining 2 QC criteria: (1) selecting the dominated health state in at least 2 out of 3 of the dominant pairs (dominant pairs violations) and/or (2) spending ≤ 150 seconds to complete all 18 DCE tasks (speeding violations).³⁵ Responses with either QC violation were excluded from the analytic data set. Additionally, to prevent fraudulent, automated survey responses from bots, Computer Automated Public Turing Test to Tell Computer and Humans Apart tasks were implemented.

Statistical Analysis

Adult and adolescent DCE responses without QC violations were analyzed using LC models specifying various numbers of classes using the `llogit2` command in Stata.^{5,11,36} Each LC model consisted of a main effect specification of 10 incremental dummy variables, in which estimated coefficients represented movement between consecutive response levels within each of the 5 dimensions. Model diagnostics included the mean square error (MSE) and mean absolute error (MAE) to quantify how well modeled probabilities predicted observed choice probabilities. Goodness of fit for each model was assessed by the Bayesian Information Criteria (BIC). Models specifying between 2 to 10 number of classes were explored. The best-fitting (lowest BIC) model was chosen as the final model for further analysis.

Across each class of the selected LC model, incremental dummy variable coefficients, sum of coefficients (within-class scale), percent of the within-class scale for each health dimension, and proportion of respondents belonging to each class (class share) were determined. The probabilities of belonging to each class were reported for each respondent and a ≥ 0.5 probability threshold was selected to assign respondents to one class, such that respondents were assigned to a specific class if they had a ≥ 0.5 probability of belonging to that class. The contribution of each class to a single estimate of latent-scale coefficients would be based on a combination of the class share and magnitude of the within-class scale, such that a class representing a greater proportion of respondents (higher class share) and/or representing a greater relative magnitude of coefficients (higher within-class scale) would contribute a greater weight. A single metric, scale-adjusted class share (SACS), was calculated for each class, combining class share percentages adjusted by the within-class scale (the relative magnitude of coefficients within each class):

$$SACS = \left(\frac{\sum \text{within class coefficients}}{\sum \text{coefficients across all classes}} \right) \times \% \text{ Class Share}$$

Classes with lower SACS would provide less weight, or contribution, to the overall estimated model coefficients as a function of (1) low number of respondents in that class, (2) low magnitude of coefficients, indicating indifferent preferences, or (3) a combination of both factors. The separate contribution of adult and adolescent respondents to the model was determined by accounting for both the proportion of adults and adolescents within each class and their individual contribution to the within-class

scale. The separate scale contribution for adults and adolescents was estimated by separate weighted conditional logit models for each class of the LC model (using the probabilities of each respondent belonging to each class as weights). To account for the unequal sample size of adults and adolescents, we performed an adjusted analysis considering an increased class share of adolescent respondents by a factor of 2.1 (dividing the number of adult respondents by the number of adolescent respondents; 1529 adults/714 adolescents). Analyses were conducted in Stata MP version 14 (StataCorp).

Results

Sample

A total of 1143 adolescents and 2573 adults initially completed the DCE survey (see Appendix Table 1 in Supplemental Materials found at <https://doi.org/10.1016/j.jval.2025.01.020>). Compared with adults, adolescents showed a greater percentage of indicating a preference for a dominated health state in the dominant pairs tasks. Upon applying the specified QC violation criteria, 429 (37.5%) adolescent and 1044 (40.6%) adult participants were excluded. Although similar proportions of adolescents and adults were excluded for QC violations overall, the types of QC violations differed between groups. Compared with adults, a significantly greater percentage of adolescents were excluded for preferring at least two-thirds of the dominated health states (14.2% vs 4.2%; $P < .01$), whereas significantly fewer adolescents were excluded for spending ≤ 150 seconds on DCE tasks (18.4% vs 29.5%; $P < .01$).

The final analytic DCE data set consisted of 714 adolescent and 1529 adult respondents (Table 1). Most adolescent participants (60.2%) were male, whereas more adult participants (54.2%) were female. Among adult participants, 51.1% were between 35 and 64 years old, and 15.4% of participants were ages 18 to 34. In response to feedback questions at the end of the DCE tasks, the majority of adolescent and adult participants strongly agreed or agreed that the questions were easy to understand and that they could tell the difference between the presented health states.

Modeling Results

LC models with 2 to 6 classes had lower MSE, MAE, and BIC with increasing number of classes (Fig. 1). The LC model with 6 classes had a similar MSE and MAE and lower BIC compared with the LC model with 7 classes and therefore was selected as the best-fitting model for further analysis.

LC Descriptions

For the selected LC model with 6 classes, classes 1 to 6 were described in terms of coefficients for incremental dummy variables, sum of coefficients (the within-class scale), and percent class share, further broken down by adolescent and adult contribution (Table 2).

Class 1 comprised 4.8% of the total respondents (representing 12% of adults, 25% of adolescents) (Table 2). Members of this class showed strong preferences related to mobility, indicated by the large discrepancy in magnitude of coefficients for problems with mobility (76% of within-class scale), compared with coefficients for all other dimensions. Class 2 comprised 10.6% of respondents (16% of adults, 18% of adolescents) who prioritized the worried, sad, or unhappy dimension (47% of within-class scale) and mobility (23% of the within-class scale). Class 3 comprised 12.4% of respondents (17% of adults, 16% of adolescents), who placed the least priority on mobility (12% of within-class scale) and similar preferences for all other dimensions. Overall, respondents in

Table 1. Adolescent and adult respondent characteristics.

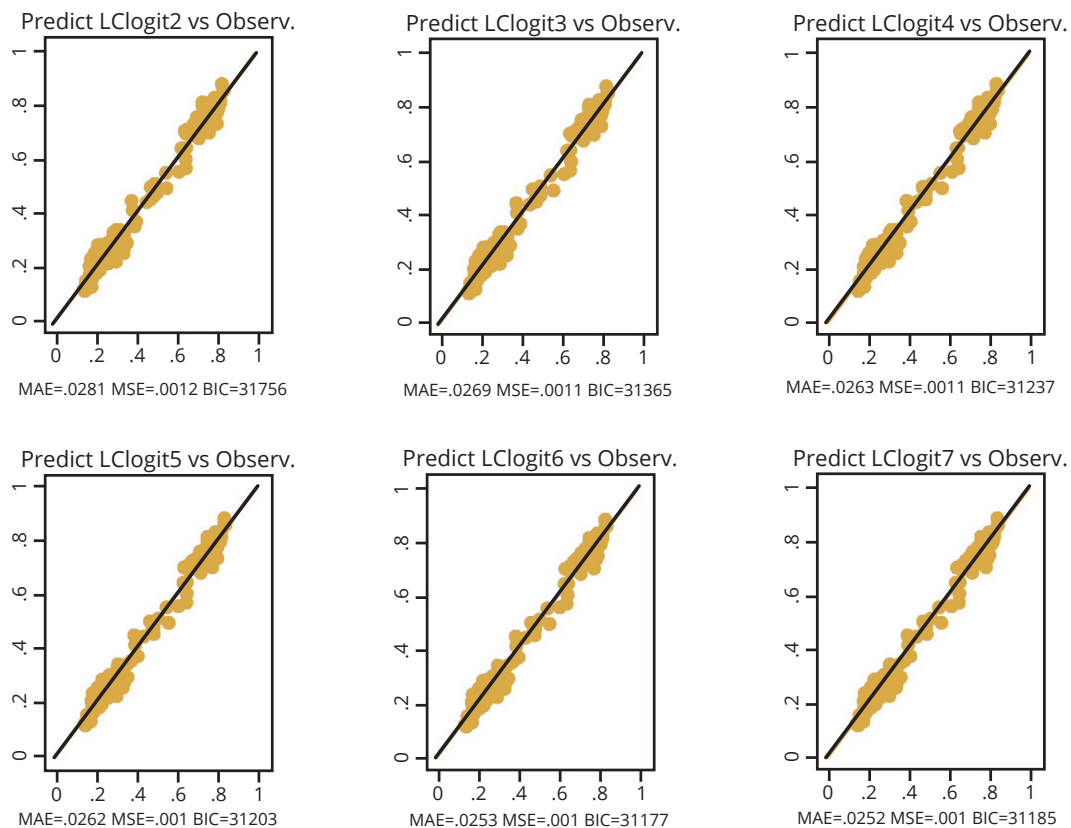
		Adolescents (<i>n</i> = 714)	Adults (<i>n</i> = 1529)
Background			
Age	Mean (SD) (min, max)	15.0 (1.81) (11,17)	53.4 (16.9) (18,93)
Age category <i>n</i> (%)	11-14	325 (45.5)	-
	15-17	389 (54.5)	-
	18-34	-	234 (15.4)
	35-64	-	781 (51.1)
	≥65	-	514 (33.6)
Gender <i>n</i> (%)	Female	283 (39.6)	828 (54.2)
	Male	430 (60.2)	698 (45.7)
	Other	1 (0.1)	3 (0.2)
Race <i>n</i> (%)	White	621 (87.0)	1252 (81.9)
	Black	54 (7.6)	168 (11.0)
	Asian	27 (3.8)	56 (3.7)
	Other	12 (1.7)	53 (3.5)
Ethnicity <i>n</i> (%)	Hispanic or Latino	76 (10.6)	150 (9.8)
Self-reported health			
Mobility <i>n</i> (%)	No problems	630 (88.2)	1246 (81.5)
	Some problems	44 (6.2)	231 (15.1)
	A lot of problems	40 (5.6)	52 (3.4)
Looking after myself <i>n</i> (%)	No problems	637 (89.2)	1436 (93.9)
	Some problems	42 (5.9)	72 (4.7)
	A lot of problems	35 (5.0)	21 (1.4)
Doing usual activities <i>n</i> (%)	No problems	619 (86.7)	1283 (83.9)
	Some problems	54 (7.6)	210 (13.7)
	A lot of problems	41 (5.7)	36 (2.4)
Having pain or discomfort <i>n</i> (%)	No	575 (80.5)	811 (53.0)
	Some	96 (13.5)	636 (41.6)
	A lot	43 (6.0)	82 (5.4)
Feeling worried, sad, or unhappy <i>n</i> (%)	Not	503 (70.5)	953 (62.3)
	A bit	168 (23.5)	466 (30.5)
	Very	43 (6.0)	110 (7.2)
EQ-VAS	Mean (Std)	88.2 (12.6)	80.2 (16.0)
Feedback			
I found it easy to understand the questions I was asked <i>n</i> (%)	Strongly agree	435 (60.9)	714 (46.7)
	Agree	201 (28.2)	549 (35.9)
	Neutral	42 (5.9)	157 (10.3)
	Disagree	28 (3.9)	83 (5.4)
	Strongly disagree	8 (1.1)	26 (1.7)
I found it easy to tell the difference between the health states I was asked to think about <i>n</i> (%)	Strongly agree	361 (50.6)	617 (40.4)
	Agree	255 (35.7)	606 (39.6)
	Neutral	65 (9.1)	191 (12.5)
	Disagree	21 (2.9)	95 (6.2)
	Strongly disagree	12 (1.7)	20 (1.3)
I found it difficult to decide on my answers to the questions <i>n</i> (%)	Strongly agree	153 (21.4)	175 (11.5)
	Agree	156 (21.9)	404 (26.4)
	Neutral	143 (20.0)	363 (23.7)
	Disagree	124 (17.4)	359 (23.5)
	Strongly disagree	138 (19.3)	228 (14.9)

SD indicates standard deviation; VAS, visual analog scale.

classes 1, 2, and 3 had similar magnitudes of within-class scales. Class 4 comprised 31.7% of respondents (22% of adults, 7% of adolescents) who had the strongest preferences overall, indicated by the largest magnitude within-class scale of -16.66 . These respondents placed most importance on worried, sad, or unhappy (30% of within-class scale) and pain/discomfort (30% of within-class scale) dimensions. Class 5 comprised 26.5% of respondents

with largely indifferent preferences (smallest magnitude of within-class scale of -1.06), consistent with a garbage class. Finally, class 6 comprised 14.1% of respondents (20% of adults and 11% of adolescents) with strong preferences (within-class scale of -15.66), particularly for mobility (23% of within-class scale) and pain/discomfort (34% of within-class scale). There were no classes unique to only adult or adolescent respondents.

Figure 1. Comparison of latent class models. Predicted vs observed probabilities, and associated model-fit statistics, for latent class models specifying between 2 and 7 latent classes.



BIC indicates Bayesian information criterion; LClogitN, latent class model with N number of classes; MAE, mean absolute error; MSE, mean square error; Observ., observed probabilities; Predict, model predicted probabilities.

In summary, classes ranked in order of class share are class 4 (31.7%), class 5 (26.5%), class 6 (14.1%), class 3 (12.4%), class 2 (10.6%), and class 1 (4.8%) and ranked in order of within-class scales, suggesting strength of preferences, are class 4 (−16.7), class 6 (−15.2), class 3 (−7.6), class 1 (−7.3), class 2 (−6.8), and class 5 (−1.1).

Examining SACS Contributions for Adolescents and Adults

SACS indicates the overall model contribution for each class, accounting for both class share and within-class scales (Table 2). By extension, the contribution of adolescents and adults to a combined model would be driven by their representation within each class according to their SACS. For example, although class 5 had the second highest class share, it had the lowest within-class scale, suggestive of a garbage class of relatively uninformative preferences. Therefore, the SACS for class 5 was the lowest of all classes at 2.9%. Although a greater proportion of adolescents were represented by class 5 compared with adults (11% of adults, 21% of adolescents, adjusting the class share for adolescents by a factor of 2.1 to account for their reduced sample size), the scale contribution of adolescents was smaller than that of adults, resulting in approximately equal SACS from adolescents and adults (Table 3). Class 4 had the highest class share and within-class scale, resulting in the largest SACS of 54.3%. Compared with other classes, class 4 would most greatly affect estimated model coefficients. Notably, class 4 also showed the greatest discrepancy in adolescent and adult representation. Although class 4 showed the

greatest class share among adults (26%), it showed the lowest adjusted class share among adolescents (9%). Accordingly, of the 54.3% SACS of class 4, 41.0% was attributed to adults, and 13.3% was attributed to adolescents (Table 3). A similar trend was observed with class 6, with a SACS of 22.0% (13.3% contribution from adults and 8.7% adjusted contribution from adolescents). Only class 1, which had the second lowest SACS of 3.6%, had a greater contribution from adolescents (2.6%) than adults (1.0%).

Accounting for the total contribution of adolescents vs adults to each SACS indicates that the LC model would be weighted with 78.7% contribution from adults and 21.3% contribution from adolescents (Table 2). When adjusting the class share for adolescents by a factor of 2.1 to approximate an equal sample size of adults and adolescents, the model contribution would be 65.0% from adults and 35.0% from adolescents (Table 3).

Discussion

This exploration of LCA demonstrated merits as a quantitative approach to understand the distribution of adult and adolescent preferences into distinct classes and determine their relative weighted contribution to a latent-scale model. LCA revealed insights about adult and adolescent prioritization of dimensions and strength of preferences based on their relative degree of representation within each class. Importantly, all classes included both adult and adolescent respondents, although a greater proportion of adolescents tended to belong to classes prioritizing mobility,

Table 2. Results of the latent class model with 6 classes—contribution from adolescents and adults.

Dimension	Response level transition	Coefficients (incremental dummy)						Totals
		Class 1	Class 2	Class 3	Class 4	Class 5	Class 6	
Mobility	Level 1 to 2	−3.01	−0.67	−0.11	−0.76	−0.25	−0.96	
	Level 2 to 3	−2.51	−0.89	−0.83	−1.06	−0.27	−2.51	
	Total of within-class scale	76%	23%	12%	11%	50%	23%	
Looking after myself	Level 1 to 2	−0.92	−0.10	−0.55	−0.54	−0.09	−0.24	
	Level 2 to 3	−0.41	−0.44	−1.30	−1.66	−0.08	−1.28	
	Total of within-class scale	18%	8%	24%	13%	16%	10%	
Doing usual activities	Level 1 to 2	−0.20	−0.17	−0.70	−1.09	−0.05	−0.73	
	Level 2 to 3	0.09	−0.31	−0.98	−1.56	−0.05	−1.84	
	Total of within-class scale	2%	7%	22%	16%	10%	17%	
Having pain or discomfort	Level 1 to 2	0.17	−0.25	−0.72	−2.34	0.02	−1.62	
	Level 2 to 3	−0.05	−0.78	−1.05	−2.66	−0.20	−3.58	
	Total of within-class scale	−2%	15%	23%	30%	17%	34%	
Feeling worried, sad, or unhappy	Level 1 to 2	−0.50	−0.99	−0.51	−1.66	0.06	−0.47	
	Level 2 to 3	0.03	−2.16	−0.86	−3.33	−0.13	−1.92	
	Total of within-class scale	6%	47%	18%	30%	7%	16%	
Sum of coefficients (within-class scale contribution)		−7.32	−6.77	−7.61	−16.66	−1.06	−15.16	−54.57
Adult		−7.18	−6.88	−7.82	−17.14	−1.50	−15.40	
Adolescent		−7.44	−6.62	−7.28	−14.62	−0.65	−14.69	
Class share contribution		4.8%	10.6%	12.4%	31.7%	26.5%	14.1%	100%
Adult		12%	16%	17%	22%	13%	20%	100%
Adolescent		25%	18%	16%	7%	23%	11%	100%
Scale-adjusted class share (SACS) contribution		3.6%	7.4%	9.7%	54.4%	2.9%	22.0%	100%
Adult		1.6%	4.6%	6.6%	47.1%	2.0%	16.8%	78.7%
Adolescent		2.0%	2.8%	3.1%	7.3%	0.9%	5.2%	21.3%

whereas more adults prioritized pain or discomfort and feeling worried, sad, or unhappy. Across classes, the notable differences in the magnitude of scales of their coefficients meant that an LC model would have greater representation from classes with the greater magnitude of scale and/or greater class share, described by the SACS metric. Despite accounting for approximately one-third of the sample, adolescents contributed only 21% of the weight to the LC model. Considering equal sample sizes of adults and adolescents, adolescents would disproportionately contribute only 35% of the total weight of the LC model.

An LCA approach revealed that relative representation of preferences in a model is not necessarily proportionate to the sample size of included subgroups. Stronger preferences for

minimizing problems in certain health dimensions are more informative in DCE tasks and are weighted more greatly in an LC model. Thus, a tendency toward less strong preferences for any specific dimension of health resulted in disproportionately lower contribution from adolescents compared with adults in a combined model. This discrepancy is not inherently problematic. Previous studies have oversampled distinct subgroups and reweighted responses in proportion to representation in the general population. Such an approach was used in the US valuation of the EQ-5D-3L to be sufficiently powered to examine whether there were racial/ethnic differences in health-state valuation.³⁷ The extent to which sociodemographic groups meaningfully affect a value set is not usually considered, although

Table 3. Contribution from adolescents and adults to the latent class model with 6 classes (assuming equal sample size of adults and adolescents).

	Class 1	Class 2	Class 3	Class 4	Class 5	Class 6	Totals
Scale contribution							
Adult	−7.18	−6.88	−7.82	−17.14	−1.50	−15.40	
Adolescent	−7.44	−6.62	−7.28	−14.62	−0.65	−14.69	
Adjusted* class-share contribution							
Adult	10%	15%	17%	26%	11%	21%	100%
Adolescent	23%	18%	16%	9%	21%	13%	100%
Adjusted* scale-adjusted class share (SACS) contribution							
Adult	1.0%	3.2%	4.9%	41.0%	1.5%	13.3%	65.0%
Adolescent	2.6%	4.1%	4.8%	13.3%	1.4%	8.7%	35.0%

*The class share of adolescents was multiplied by a factor of 2.1 to account for the unequal sample size of adult ($n = 1529$) and adolescent ($n = 714$) respondents.

this becomes increasingly important when (1) a certain level of contribution is expected from distinct subgroups and (2) there is a disparity in the informativity of their preferences. Decision makers may assume that an increased sample size will result in proportionately increased representation in terms of information to a value set. Our analysis indicates that this is an erroneous assumption. For example, this analysis suggests a requirement of approximately twice as many adolescents as adults to achieve a model with equal contribution in terms of information. In thinking about the level of representation of subgroups to a model, there needs to be clear communication about which type of representation is expected, namely inclusion in the valuation process or contribution to the level of information represented by their preferences. For example, we oversampled adolescents ages 11 to 17 relative to their proportion of the US population, but their disproportionately lower contribution to a model may be lacking the degree of representation expected by stakeholders.

Inclusion of adolescents, as recommended by US stakeholders, is consistent with providing younger populations a voice and representation in matters that may affect decisions related to their health.^{15,18,19} Although there is considerable interest in engaging children in preference-based studies of health, there remain conceptual and empirical issues to their inclusion.^{13-15,33,38} Such issues range from how and why to include children to what ages are acceptable and sufficiently informed to participate. Engaging children in health preference research can be more technically and logistically complex because of potential ethical issues and more intensive resource requirements to recruit children. To strike a balance between the desires of stakeholders to incorporate children for the purpose of inclusivity and the practical value when considering the additional complexity and cost, more research on the informativeness of preferences of children is needed to guide researchers. Inclusion of adolescents in valuation remains a normative decision and pursued in this study based on the recommendation of US stakeholders, although this was a small sample ($n = 14$), and consensus on their inclusion was not reached. This is not only relevant to the EQ-5D-Y-3L but also for the valuation of other pediatric HRQoL instruments in which adolescent inclusion may be desirable, such as the CHU9D and PedsQL.^{39,40} In the context of the US valuation of the EQ-5D-Y-3L, a final value set would further be derived by anchoring latent DCE values to cTTO-derived values obtained only from adults, further polarizing the representation of adults compared with adolescents.

This study relied on some assumptions with associated limitations. In terms of the DCE tasks, different framing perspectives were used between adolescent (self) and adult (10-year-old-child) DCE tasks as done in previous studies.^{13,14,41,42} The degree to which the difference in framing perspective of DCE tasks between adolescent and adult samples contributed to the observed class memberships and model contribution cannot be disentangled and may be a constraint of combining their DCE responses in this way. In terms of the analytic approach, class membership in LCA is probabilistic; therefore, determination of an individual respondent's class membership was based on a chosen probability threshold of ≥ 0.5 of belonging to that class. Other probability thresholds could have been chosen, resulting in different assignments of class membership, although with the trade-off of over or underrepresenting individuals in each class. Further, separately modeling scale-classes and taste-classes or selecting different model-fit criteria may have resulted in the selection of an LC model specifying a different number of classes.³² In the 6-class model, there was an observed representation of adolescent and adult respondents across all classes. Regardless of the number of classes, their proportional representation within the most

informative classes would unlikely change. Thus, even with a different number of classes, we would anticipate a similar contribution from adults and adolescents. The choice of the model with 6 classes was optimal compared with other model choices in terms of information provided and fit, consistent with the selection criteria used in another study using a similar LCA approach.⁵ One limitation of the SACS metric is that it would naturally be lower for adolescents because they represent less of the total sample compared with adults. To adjust for this, we considered how accounting for the difference in sample size would change adolescent representation in the LC model, although with the assumption that doubling the sample size of adolescents would result in the same observed distribution and scale contribution to each class. Lastly, in terms of sampling, quota sampling was not strictly enforced to balance feasibility of recruitment, leading to underrepresentation of some groups, particularly among adolescents.

In terms of data quality, although we implemented standard QC criteria and self-reported feedback questions,^{2,35} these are imperfect indicators that do not explain respondent behavior. For example, selecting a dominated health state in dominant pairs task may indicate a lack of comprehension or inattention; speeding through tasks may indicate lack of engagement or, conversely, a thorough understanding, enabling easy completion of the tasks. Among all respondents, adolescents tended to prefer a dominated health state with a greater frequency than adults in the 3 dominant pairs tasks. Consequently, more adolescents were excluded based on the QC criteria of $\geq 2/3$ dominant pairs violations. However, this also means that more adolescents with $1/3$ dominant pairs violations were included in the analytic sample. This inattention and/or lack of comprehension from adolescents included in the analytic sample could indicate inexperience with described health states and may have potentially resulted in their preferences being less informative to the model. Among respondents passing QC and included in the analytic sample, few adolescents or adults reported that DCE tasks were difficult to understand. Nevertheless, this contrast in behavior (dominant pair violations) and self-reported understanding could be attributed to social-desirability bias. Particularly from adolescents, data quality could have been improved using stricter QC criteria (eg, excluding respondents with ≥ 1 dominant pair violation, as done in some studies¹⁴), although with the trade-off of excluding a greater proportion of adolescents compared with adults and further reducing their representation in a final model. Rather, structuring surveys with practice DCE tasks and/or conducting interviewer-assisted surveys may be potential tactics to further improve understanding and engagement with tasks, ultimately minimizing QC violations and maximizing the information obtained. Future research could characterize respondents belonging to each class to identify other respondent characteristics beyond age group associated with indifferent preferences.

Although adolescents have demonstrated capability in completing DCE tasks and there is an increasing interest in their inclusion in valuing HRQoL, this study demonstrates that the inclusion of adolescents in a valuation study warrants additional consideration to make a meaningful contribution to a latent-scale model. By extension, although researchers may take care in sampling respondents from certain subgroups, such as racial/ethnic groups, in an attempt to ensure their representation in a value set, LCA may reveal a disproportionate contribution when accounting for the degree of informativity of their preferences. The methods used in this research could be adapted as a means to more thoroughly vet the representativeness of a DCE-based value set and to understand the strengths and patterns of preferences among other subgroups of a population.

Conclusions

When valuing EQ-5D-Y-3L health states, adolescents exhibited preferences that were less informative to an LC model compared with adult preferences. Findings from this study offer additional points of consideration in the ongoing discussions about sources of preferences—adults, adolescents, or both—and their level of contribution in valuation for measures of children's health. In consideration of the advantages and disadvantages, the LCA approach in this study represents a potential method to combine adolescent and adult responses and can be explored in other applications of examining and weighting preferences from intentionally sampled subgroups.

Author Disclosures

Author disclosure forms can be accessed below in the [Supplemental Material](#) section.

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